

Introduction

Supervised learning uses labelled data $(x_i, y_i)_{i=1}^n$ to learn a function f that predicts y from new x . The enterprise is defined by three choices: a **hypothesis class** (tree, linear model, neural net), a **loss function** (squared error, cross-entropy), and an **optimisation** procedure. Generalisation – performance on unseen data – is the goal, and the bias-variance decomposition is the classical lens for understanding why models fail.

Prerequisites

Probability, linear algebra, basic statistics.

Theory

Empirical risk minimisation (ERM): $\hat{f} = \arg \min_{f \in \mathcal{F}} \frac{1}{n} \sum_i L(y_i, f(x_i))$.

Loss functions: - Regression: squared loss $(y - f)^2$, absolute loss $|y - f|$. - Classification: 0/1 (intractable), log loss (cross-entropy), hinge (SVMs).

Bias-variance decomposition for squared loss:

$$\mathbb{E}[(y - \hat{f}(x))^2] = \text{Bias}^2 + \text{Variance} + \sigma_{\text{noise}}^2.$$

Flexible models have low bias, high variance; rigid models the reverse. Regularisation trades one for the other.

Assumptions

(x, y) pairs are drawn iid from some joint distribution; the deployment distribution matches the training distribution (no distribution shift).

R Implementation

```
library(tidymodels)

set.seed(2026)
n <- 300
x <- runif(n, 0, 2 * pi)
y <- sin(x) + rnorm(n, 0, 0.3)
df <- tibble(x = x, y = y)

split <- initial_split(df, prop = 0.7)
train <- training(split); test <- testing(split)

# Two models of different flexibility
m_rigid <- lm(y ~ x, data = train)
m_flex <- loess(y ~ x, data = train, span = 0.2)

rmse <- function(yhat, y) sqrt(mean((yhat - y)^2))
rmse(predict(m_rigid, test), test$y)
rmse(predict(m_flex, test), test$y)
```

Output & Results

The flexible loess model has lower test RMSE on this non-linear signal; the rigid linear model under-fits.

Interpretation

“The sinusoidal signal is captured by loess (test RMSE 0.31) but missed by linear regression (test RMSE 0.68), illustrating how bias dominates when the hypothesis class is too restrictive.”

Practical Tips

- Always report test performance, never training performance, as a generalisation estimate.
- The no-free-lunch theorem: no single model wins everywhere; match model to the data-generating process.
- Stratify classification splits to preserve class proportions.
- Use cross-validation for small-to-moderate data; a single train/test split is high-variance.
- Track multiple metrics: RMSE and MAE together are more informative than either alone.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Cross-validation, nested CV, bootstrap .632+
- Regularisation: ridge, lasso, elastic net
- Trees, random forests, gradient boosting
- Interpretability and SHAP
- Tabular neural networks with torch
- Imaging and sequence models intro
- tidymodels pipelines
- FDR, knockoffs, replication crisis